

Theory Assignment 1: Losses, Optimisers, Preprocessing, Validation and Regression

CSAI2017P: Applied Machine Learning

Units I–IV and Unit V (part) • Lecture packages L1–L10 • CO1, CO2, CO3

Instructor: Dr. Tofik Ali

Student Name: _____

Batch: _____

Due: at AS1, announced in class

SAP ID: _____

Lab quiz: AT1, in class at AS1

What this assignment is for

This is the first of two theory assignments, and it is deliberately built as your **mid-semester preparation**. Sections A to C make you rehearse Units I–IV *and the first three lectures of Unit V* in the three ways an examination asks for material: explain it, compute it by hand, and apply it in code. Section D is a full mock paper, to be attempted under examination conditions.

The mid-semester syllabus is not confirmed yet

The examination cell has not issued the notice. **Units I–IV plus Unit V as far as L10 is an expectation, not an announcement** — a well-founded one, because a paper can only examine what has been taught, and lecture packages L1–L10 are what the calendar delivers before the mid-semester point. **Unit V is covered only to L10:** regression foundations and least squares (L8), maximum likelihood and model adequacy (L9), and logistic regression (L10). Multiple linear regression (L11) and regularization (L12) are *not* examinable here. I will tell you as soon as anything is confirmed.

Nothing about that uncertainty should change what you do now. Everything in this assignment is examinable material whatever the mid-semester notice says — if not in November then in the end-semester paper, which covers all seven units. The *Mid-Semester Question Bank* released alongside this brief is organised unit by unit precisely so that it still works if the coverage turns out narrower or wider than expected; its opening pages explain what to do in each case.

How the marks work

The assignment is marked out of **50 raw**, scaled to the **1.0** submission mark. The in-class assessment test **AT1** that follows at AS1 is worth **1.5** marks — more than the submission itself. AT1 asks you about *your own* submitted work: where a number came from, why you chose one option over another, what a fragment of your own code does. If you did the assignment, AT1 takes a few lines per answer. If you did not, no amount of polish on the submission will save it. This is the design, not an ambush.

Sec.	What it tests	Choice rule	Marks
A	Concepts, Units I–V(pt)	Attempt any 5 of 16 (2 marks each)	10
B	Numericals worked by hand	Attempt any 2 of 12 (5 marks each)	10
C	Pipeline, and a regression fit	Compulsory	12
D	Mock mid-semester paper (out of 35, scaled)	Compulsory	18
Raw total			50
Scaled to the submission mark			1.0
Assessment test AT1, in class at AS1			1.5

Course outcomes assessed: **CO1, CO2, CO3**. Attempting more than the choice rule allows is not penalised; your best attempts are marked.

Why there are so many questions you do not have to answer

Sections A and B offer twenty-eight items and ask for seven. The marks are unchanged and so is the workload — but the twenty-eight between them cover every part of Units I–V(pt) that the mid-semester paper can reach, so the ones you skip are still worth reading, and the ones you choose are yours. Pick the areas you are least sure of rather than the ones you can already do; the marks are the same either way and only one of those choices helps you in the examination hall.

The **Mid-Semester Question Bank** released alongside this brief carries a further 314 questions across the same syllabus, 63 of them numerical. It is not assessed. It is the revision resource.

Instructions

- **Sections A, B and D are handwritten**, on ruled A4, scanned to a single PDF. This is on purpose — the examination is handwritten and under time pressure, and practising on a keyboard does not prepare you for that.
- **Section C is a notebook**, all cells executed, outputs visible.
- In Section B, **show every step**. A correct final number with no working earns 1 mark out of 5. A wrong final number with correct method earns most of the marks.
- Report the metric named in the question. A number with no units or context earns nothing.
- Set `random_state` everywhere in Section C. If your results are not reproducible they are not results.
- Fit every transformer inside a `Pipeline`. Nothing is fitted on data the model will be evaluated on.

What to submit

- `AML_A1_<SAPID>_written.pdf` — scanned Sections A, B and D.
- `AML_A1_<SAPID>.ipynb` — Section C, executed.
- `AML_A1_<SAPID>_report.pdf` — 2–3 pages, or a `README.md`, carrying your Section C results table and observations.

- Do not upload datasets. Cite the source and include the loading code.
- **Bring the printed or written submission to AS1.** AT1 is open-submission and closed everything else.

Get the data before you start

Section C uses anchor dataset **A3 (Statlog German Credit)**, fetched with `fetch_openml('credit-g', version=1, as_frame=True)`. It downloads once (~0.2MB) and then caches. **Run the fetch at home, on your own connection.** The campus network may block it. Full details are in **Anchor-Datasets.pdf** alongside the lab assignments.

Academic integrity

Discussing ideas with your classmates is encouraged and is how most people learn this material. Submitting someone else's code or text as your own is not. If you used a tutorial, a textbook chapter or an AI tool substantially, cite it in your report — that is honest practice, not an admission of weakness, and it costs you no marks. What does cost marks is AT1, which you will not be able to answer about work you did not do.

Section A (Concepts) — attempt any 5 of 16 ($5 \times 2 = 10$)

Answer each in 4–6 lines. Marks are for precision, not length. The sixteen are ordered by unit: **A1–A2** Unit I, **A3–A4** Unit II, **A5–A6** Unit III, **A7–A12** Unit IV, **A13–A16** Unit V. **Choose at least one from each unit** — five from one corner of the syllabus is not revision.

A1. Types of learning [2 marks]

Classify each of the following as supervised, unsupervised or reinforcement learning, and state in one line what the training signal is in each case.

- Predicting tomorrow's electricity demand from ten years of hourly meter readings.
- Grouping 50,000 unlabelled customer profiles into segments for a marketing team.
- Training a robot arm to stack blocks, where it is scored only on whether the tower stands.

A2. The three libraries [2 marks]

NumPy, Pandas and scikit-learn each own a distinct job in a Python ML workflow. State the job of each in one line. Then explain what scikit-learn's `fit`, `transform` and `predict` each do, and why `fit_transform` exists but `fit_predict` is rare.

A3. Choosing a regression loss [2 marks]

A house-price model is trained on data where about 2% of the recorded prices are data-entry errors an order of magnitude too large. Compare MSE, MAE and Huber loss for this problem. Name the one you would use, and give the property of the Huber loss that makes it a compromise between the other two.

A4. Cross-entropy variants [2 marks]

Binary, categorical and sparse categorical cross-entropy are the same idea in three packagings. For each, state (i) the number of classes it applies to, (ii) the shape and format the *labels* must be in, and (iii) the activation the model's final layer should use.

A5. Batch size and momentum [2 marks]

Explain the trade-off between full-batch, mini-batch and stochastic gradient descent in terms of gradient noise and cost per update. Then write the momentum update rule and state, in one line each, what happens when $\beta = 0$ and when β is close to 1.

A6. Adagrad, RMSprop and Adam [2 marks]

- (a) Write the update rule for each of the three.
- (b) State the one failure mode of Adagrad on a long training run, and name the single change RMSprop makes to fix it.
- (c) Name the two things Adam adds on top of RMSprop.

A7. Fitting the scaler [2 marks]

A classmate standardises the whole dataset, then splits it 80/20 and reports test accuracy. Name what has gone wrong, explain the mechanism in two lines, and state whether the reported accuracy is likely to be optimistic or pessimistic — with a reason.

A8. Imbalance [2 marks]

A fraud dataset is 2% positive. State one advantage and one danger of random over-sampling, and the same for random undersampling. Then state exactly where in a k -fold cross-validation loop the resampling must happen, and why it must not happen before the loop.

A9. Masking [2 marks]

A column of 12 sensor readings contains two obviously wrong values. The $|z| > 3$ rule flags neither of them; the $1.5 \times \text{IQR}$ fence flags both.

- (a) Name the effect and explain the mechanism in two lines.
- (b) State the breakdown point of each of the two rules.
- (c) Name a third rule with a higher breakdown point than either.

A10. The four scalers [2 marks]

Name `StandardScaler`, `MinMaxScaler`, `RobustScaler` and `MaxAbsScaler`, write each formula, and give the output range each guarantees. State which one a single typing error destroys completely, and why that one is the most dangerous despite producing output that looks perfectly correct.

A11. Encoding [2 marks]

A colour column holds `red`, `green`, `blue`. A classmate applies `OrdinalEncoder` because “it makes fewer columns”.

- (a) State the assertion this makes about the variable that is not true.
- (b) State which model families believe that assertion and which ignore it.
- (c) Give the encoding that should have been used and the number of columns it produces, with and without `drop='first'`.

A12. PCA [2 marks]

“PCA reduces dimensionality by keeping the most important features.” State precisely

what is wrong with that sentence, say what PCA actually keeps, and give one reason a hospital wanting to order fewer blood tests would not be helped by it at all.

A13. The least-squares line [2 marks]

- (a) State exactly which deviations least squares minimises — be precise about the word “vertical”.
- (b) Give the two normal equations, and the estimators b_1 and b_0 that follow.
- (c) State the four properties of the fitted line, and say which one requires the model to have an intercept.

A14. Maximum likelihood [2 marks]

- (a) State the noise assumption under which least squares *is* the maximum likelihood estimator, in one sentence.
- (b) Explain why the value of σ^2 cannot affect which β_0, β_1 maximise the likelihood.
- (c) State what maximum likelihood minimises instead if the noise is Laplace rather than Gaussian.

A15. R^2 and what it hides [2 marks]

A classmate reports $R^2 = 0.87$ on the training set and concludes the model is good.

- (a) Give two distinct things wrong with that inference.
- (b) State why R^2 can never fall when a predictor is added, in one sentence.
- (c) State the one plot that should accompany every reported R^2 , and what it would reveal that the number cannot.

A16. Odds, log-odds and the sigmoid [2 marks]

- (a) Give the range of a probability, of the odds, and of the log-odds. State which one a linear expression $w^\top x + b$ can safely produce, and why that settles where the model is built.
- (b) Invert $\ln(p/(1-p)) = z$ to obtain $p = \sigma(z)$, showing the steps.
- (c) State what e^{β_j} is a change in. *Answering “probability” scores zero for this part.*

Section B (Numericals) — attempt any 2 of 12 ($2 \times 5 = 10$)

Work by hand. Show every step. You may check your answer with Python afterwards, but the working is what is marked. Give final values to four decimal places.

B1. Three losses on one set of errors [5 marks]

A regression model produces these predictions:

y	3.0	-0.5	2.0	7.0	4.5
$\hat{y}$	2.5	0.0	2.0	8.0	1.5

- (a) Compute the MSE and the MAE. [2 marks]
- (b) Compute the Huber loss with $\delta = 1$, showing which branch of the piecewise definition each of the five points falls into. [2 marks]
- (c) The fifth point has error 3. Compute what *fraction* of the total MSE and of the total Huber loss that single point contributes, and say in two lines what this tells you about robustness. [1 marks]

B2. Gradient descent, with and without momentum [5 marks]

Fit $\hat{y} = wx$ (no intercept) by minimising $L(w) = \frac{1}{n} \sum_i (y_i - wx_i)^2$ on the three points $(1, 2)$, $(2, 4)$, $(3, 5)$. Start at $w_0 = 0$ with learning rate $\eta = 0.1$.

- (a) Derive dL/dw and evaluate it at w_0 . [1 marks]
- (b) Perform two full-batch gradient descent updates. Report w_1 and w_2 . [1.5 marks]
- (c) The least-squares optimum is $w^* = \sum x_i y_i / \sum x_i^2$. Compute it and state how close w_2 is. [1 marks]
- (d) Now redo the same two steps with momentum, $v_t = \beta v_{t-1} + g_t$ and $w_t = w_{t-1} - \eta v_t$, with $\beta = 0.9$ and $v_0 = 0$. Report w_1 and w_2 , and explain in three lines what momentum has done here relative to w^* . [1.5 marks]

B3. Adam against RMSprop against SGD [5 marks]

A single parameter starts at $w_0 = 1.0$. Two successive mini-batches give gradients $g_1 = 0.5$ and $g_2 = 0.3$. Use $\eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\rho = 0.9$, $\varepsilon = 10^{-8}$.

- (a) Two steps of plain SGD. [0.5 marks]
- (b) Two steps of Adam. Tabulate m_t , v_t , $\hat{m}_t$, $\hat{v}_t$ and w_t at each step. [2.5 marks]
- (c) Two steps of RMSprop, $v_t = \rho v_{t-1} + (1 - \rho)g_t^2$. [1 marks]
- (d) Adam's first step is almost exactly $-\eta$ regardless of the size of g_1 . Show why from your numbers, and explain in three lines what the bias-correction terms $1 - \beta_1^t$ and $1 - \beta_2^t$ are for. [1 marks]

B4. Outliers, scaling and the masking effect [5 marks]

A feature column holds the values 12, 15, 14, 10, 13, 60, 11, 14.

- (a) Compute the mean and the population standard deviation (ddof = 0, as **StandardScaler** uses). [1 marks]
- (b) Compute Q_1 , Q_3 , the IQR and the $1.5 \times \text{IQR}$ fences. Which values are flagged as outliers? [1.5 marks]
- (c) Compute the z -score of the value 60. Using the common $|z| > 3$ rule, is it flagged? [1 marks]
- (d) You have two methods disagreeing about the same point. Explain the mechanism — what the value 60 does to the statistic used to judge it — and name which of the two rules you would trust here and why. [1 marks]
- (e) Compute the min-max scaled value of 12 with the outlier present. State in two lines what this shows about min-max scaling on outlier-contaminated data. [0.5 marks]

B5. Cross-entropy, three ways**[5 marks]**

- (a) Four emails have labels $y = (1, 0, 1, 0)$ and predicted probabilities of class 1 $p = (0.8, 0.3, 0.6, 0.1)$. Write the probability each assigns to its *correct* class, then compute the binary cross-entropy. **[2 marks]**
- (b) The third email's probability now falls to 0.02. Recompute the BCE and state the increase as a percentage. **[1 marks]**
- (c) A three-class model outputs $P = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}$ with true classes 0, 1, 2. Compute the categorical cross-entropy from one-hot labels and the sparse categorical cross-entropy from integer labels. **[1.5 marks]**
- (d) Your two answers in (c) are identical. Explain why the sparse form exists at all, with a memory argument for $K = 10,000$ classes and a batch of 256. **[0.5 marks]**

B6. Imputation and the shrinkage it hides**[5 marks]**

Eight students sat a test; two scripts were lost. The six recorded marks are 48, 55, 60, 67, 72, 78.

- (a) Compute the mean, median and standard deviation of the six observed marks. State clearly which divisor you used. **[1 marks]**
- (b) Impute the two missing marks with the mean of the observed. Compute the mean and the population standard deviation of all eight. **[1 marks]**
- (c) Repeat with median imputation. **[1 marks]**
- (d) Compute $\sqrt{m/n}$, where m is the number observed. Compare it with the ratio of your imputed standard deviation to the observed one, and derive why the two agree. **[1.5 marks]**
- (e) State in two lines what this means for any standard error or correlation computed after mean imputation. **[0.5 marks]**

B7. PCA by hand**[5 marks]**

Five points: (1, 1), (2, 3), (3, 2), (4, 5), (5, 4).

- (a) Compute the mean and write the centred matrix. **[0.5 marks]**
- (b) Compute the 2×2 covariance matrix using the divisor $n - 1$. **[1 marks]**
- (c) Find both eigenvalues. Verify that they sum to the trace, and give the proportion of variance each component explains. **[1.5 marks]**
- (d) Give the first eigenvector, project the five centred points onto it, and verify that the variance of the five scores equals λ_1 . **[1 marks]**
- (e) Reconstruct the five points from one component and compute the total squared reconstruction error. Verify it equals $(n - 1)\lambda_2$. **[1 marks]**

B8. The confusion matrix and the metric that lies**[5 marks]**

A screening model on 10,000 patients returns $TN = 9900$, $FP = 60$, $FN = 30$, $TP = 10$.

- (a) Compute accuracy, precision, recall, F_1 and balanced accuracy. [2.5 marks]
- (b) Compute the accuracy of a model that predicts “negative” for every patient. [0.5 marks]
- (c) State which of the two models scores higher on accuracy, and what that tells you about accuracy as a metric here. [1 marks]
- (d) Name the metric you would put in front of a clinical audit committee, and justify it in three lines in terms of the cost of a false negative. [1 marks]

B9. Least squares by hand [5 marks]

Fit a least-squares line to $x = (0, 1, 2, 3, 4)$, $y = (1, 2, 4, 5, 8)$.

- (a) Tabulate $x_i - \bar{x}$, $y_i - \bar{y}$, $(x_i - \bar{x})^2$ and $(x_i - \bar{x})(y_i - \bar{y})$; give S_{xx} , S_{xy} and S_{yy} . [1.5 marks]
- (b) Compute b_1 and b_0 and write the fitted line. [1 marks]
- (c) Give the fitted values and residuals, and verify *both* $\sum e_i = 0$ and $\sum x_i e_i = 0$. [1 marks]
- (d) Compute SSE, SSR and SST, verify the decomposition, and check SSR a second independent way. [1 marks]
- (e) Predict y at $x = 6$. [0.5 marks]

B10. Which line is the least-squares line? [5 marks]

Four candidates are proposed for the data of B9.

A: $0.5 + 1.7x$	B: $0.6 + 1.7x$	C: $0.6 + 1.8x$	D: $1.0 + 1.5x$
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- (a) For each, compute SSE, $\sum e_i$ and $\sum x_i e_i$. [3 marks]
- (b) Identify the least-squares line using *both* normal equations. [1 marks]
- (c) One candidate satisfies $\sum e_i = 0$ and is still not the answer. Name it, and explain in three lines why the first normal equation alone cannot pin down a line. [1 marks]

B11. R^2 , and the two estimates of σ^2 [5 marks]

Using your fit from B9 (SSE = 1.10, SST = 30, $n = 5$):

- (a) Compute R^2 from the definition, then verify it two further independent ways. [1.5 marks]
- (b) Compute r and confirm $r^2 = R^2$. [0.5 marks]
- (c) Compute $\hat{\sigma}_{\text{ML}}^2 = \text{SSE}/n$, the unbiased $s^2 = \text{SSE}/(n - 2)$, and s . [1 marks]
- (d) Give the bias factor and the percentage by which the maximum likelihood estimate understates σ^2 in expectation. [1 marks]
- (e) Explain *why* the divisor is $n - 2$. [1 marks]
An answer that says “because we divide by n instead of $n - 2$ ” restates the formula and earns nothing.

B12. Logistic regression: one gradient step by hand [5 marks]

Take $X = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, $y = (0, 0, 1, 1)$, $w = (0, 0)$ and $\eta = 0.4$.

- Compute z , then p , then the log-likelihood. Express the per-row negative log-likelihood as a familiar constant and say what it is the baseline for. [1 marks]
- Compute $p - y$ and the gradient $X^\top(p - y)$, showing both components. [1.5 marks]
- Perform one gradient step and give the new w . [0.5 marks]
- Compute the new z , the new p and the new log-likelihood, and state the improvement. [1 marks]
- The intercept component of the gradient is exactly zero. Explain why, and state what would have to change for it not to be. [1 marks]

Section C (Pipeline and regression) — compulsory (12)

Dataset: A3 (Statlog German Credit), 1000 rows, 20 mixed features, binary target `class` $\in \{good, bad\}$. Roughly 70/30 imbalanced.

C1. Build the pipeline [4 marks]

- Load A3. Report the number of rows, the number of numeric and categorical features separately, and the class balance.
- Build a `ColumnTransformer` that median-imputes and standardises the numeric columns and mode-imputes and one-hot encodes the categorical ones. Use `OneHotEncoder(handle_unknown='ignore')` and say in one line why that argument matters.
- Wrap it with `LogisticRegression(max_iter=2000)` in a `Pipeline`. Split 80/20 with `stratify=y`, `random_state=42`.
- Report the number of columns *after* encoding, the test accuracy and the test ROC-AUC.
- Run `StratifiedKFold(5, shuffle=True, random_state=42)` cross-validation on the training portion and report mean $\pm$ standard deviation of ROC-AUC.

C2. Make leakage visible [3 marks]

This part uses *pure noise*, so anything above chance is leakage and nothing else.

- Generate `X = rng.normal(size=(200, 5000))` and `y = rng.randint(0, 2, 200)` with `rng = np.random.RandomState(0)`. There is, by construction, no relationship between `X` and `y`.
- The wrong way.* Apply `SelectKBest(f_classif, k=20)` to the *whole* of `X` and `y`, then cross-validate a logistic regression on the selected 20 columns with `StratifiedKFold(5, shuffle=True, random_state=0)`. Report the mean accuracy.
- The right way.* Put the same `SelectKBest` inside a `Pipeline` with the classifier and cross-validate the pipeline on the full `X`. Report the mean accuracy.

- (d) Explain in 5–8 lines why (b) scores what it scores. Your answer must name what the selector saw that it should not have, and say which of the two numbers an honest report would quote.

C3. Imbalance and the metric you report [3 marks]

Back to A3, treating *bad* credit as the positive class.

- (a) Report accuracy, precision, recall and ROC–AUC for your C1 pipeline on the test set.
- (b) Refit with `class_weight='balanced'` and report the same four numbers.
- (c) Present both rows in one table. Three of the four numbers move in an interesting way and one barely moves. Say which, and explain in 4–6 lines why ROC–AUC behaves differently from accuracy under a change of class weighting.
- (d) A bank asks which of the two models to deploy. Answer in three lines, and your answer must refer to the *cost* of the two kinds of error, not only to the numbers.

C4. Regression, by hand and by library [2 marks]

No download needed — use `sklearn.datasets.load_diabetes`, which ships with *scikit-learn* and works offline. Take `bmi` as the single predictor and the disease-progression score as the target.

- (a) Split 75/25 with `random_state=0`. On the *training* rows only, compute $\bar{x}$, $\bar{y}$, S_{xx} , S_{xy} and hence b_1 and b_0 **in NumPy, from the formulae** — no `LinearRegression` in this part. [0.75 marks]
- (b) Now fit `LinearRegression()` on the same rows. Print your two coefficients beside the library's and report the absolute difference. [0.5 marks]
- (c) Report training and test RMSE, and the test R^2 . [0.5 marks]
- (d) Report the RMSE of predicting the training mean for every test row, and say in two lines what your R^2 means relative to it. [0.25 marks]

If (b) does not agree to at least six decimal places, your (a) is wrong — find it before moving on. That agreement is the whole point of the part.

Section D (Mock mid-semester paper) — compulsory (18)

Attempt this under examination conditions

90 minutes. Closed book. No computer, no calculator beyond arithmetic. Sit down, set a timer, and stop when it goes. The paper below is written to the mid-semester pattern and syllabus *as currently expected* — see the note on the first page; neither is confirmed. Marked out of **35 raw**, scaled to **18** within this assignment — the real paper is scaled the same way to its own weight. Submit your timed script with the rest of the assignment. The answer key is released after AS1; do not go looking for it first, because the whole value of this section is in finding out what you cannot yet do while there is still time to fix it.

Mock paper — Section A: attempt all ($5 \times 2 = 10$)

1. State exactly which deviations least squares minimises, write the two normal equations, and give b_1 and b_0 in terms of S_{xx} and S_{xy} . [2 marks]
2. Write the formulae for MSE and MAE. State which is more sensitive to outliers and justify it from the formula, not from memory. [2 marks]
3. A 4-class classifier outputs a softmax vector. Your labels are stored as the integers 0, 1, 2, 3. Name the correct loss function, and state what you would have to change to use categorical cross-entropy instead. [2 marks]
4. Write the SGD-with-momentum update rule and state in one line what β controls and what happens as $\beta \rightarrow 1$. [2 marks]
5. Define data leakage in one sentence and give two distinct examples of preprocessing steps that cause it if performed before the train–test split. [2 marks]

Mock paper — Section B: attempt any 3 of 4 ($3 \times 5 = 15$)

6. A model produces $y = (10, 12, 15, 9)$ and $\hat{y} = (11, 10, 20, 9)$. Compute the MSE, the MAE and the Huber loss with $\delta = 2$, showing the branch taken by each point. Comment in two lines on which loss the third point dominates. [5 marks]
7. Fit a least-squares line to $x = (0, 1, 2, 3, 4)$, $y = (1, 2, 4, 5, 8)$. Give S_{xx} , S_{xy} , b_1 and b_0 ; the fitted values and residuals; and verify *both* $\sum e_i = 0$ and $\sum x_i e_i = 0$. Compute SSE, SSR, SST and R^2 , checking SSR a second independent way. [5 marks]
8. Write the update rules for Adagrad, RMSprop and Adam. State the one failure mode of Adagrad on long training runs, name the change RMSprop makes to fix it, and name the two things Adam adds on top of RMSprop. [5 marks]
9. Take X with rows $(1, -1), (1, 0), (1, 1), (1, 2)$, $y = (0, 0, 1, 1)$, $w = (0, 0)$ and $\eta = 0.4$.
 - (a) Compute p and the log-likelihood, and express the per-row negative log-likelihood as a familiar constant. [1.5 marks]
 - (b) Compute the gradient $X^\top(p - y)$, showing both components, and perform one gradient step. [2 marks]
 - (c) Give the new p and the new log-likelihood. [1 marks]
 - (d) Explain why the intercept component of the gradient is exactly zero here. [0.5 marks]

Mock paper — Section C: compulsory ($1 \times 10 = 10$)

10. A hospital gives you 5000 patient records: 12 numeric features (two of them with about 4% missing values and visible outliers), 3 categorical features, and a binary target that is 8% positive. You must build and validate a model, and defend it to a clinical audit committee.
 - (a) Set out the complete preprocessing sequence, from raw table to model input, in order. For each step state *what* it does and *why* it is at that position in the order. [3 marks]
 - (b) State precisely which steps are fitted on training data only, and describe what happens inside a single fold of 5-fold stratified cross-validation — fold sizes, number of positives, and which objects are refitted. [3 marks]

- (c) You decide to oversample the minority class. State exactly where in the fold this happens, and explain what goes wrong if you oversample before the folds are cut. **[2 marks]**
 - (d) The committee asks for a single headline number. Explain why accuracy is a poor choice here, name the metric you would report instead, and justify it in terms of the clinical cost of a false negative. **[2 marks]**
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End of A1. The assessment test AT1 follows at AS1 and is worth 1.5 marks. Bring your submission.